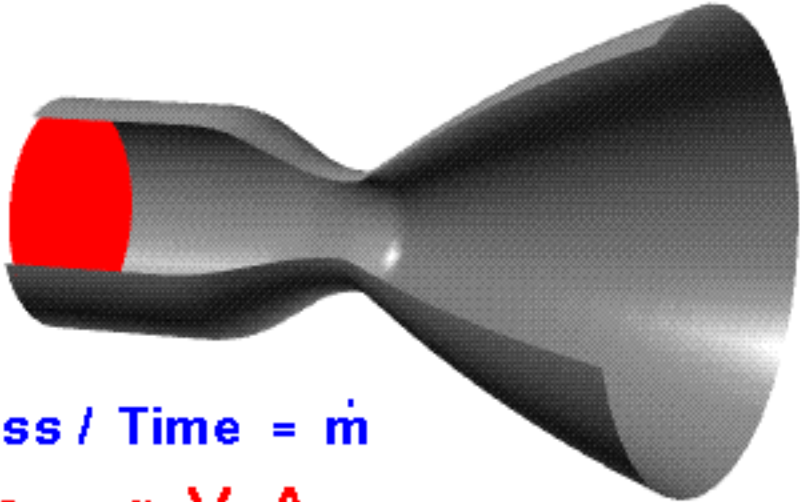
mass conservation constraint

r = Density

V = Velocity

A = Area



Mass Flow Rate = Mass / Time = $\dot{m}$

$$\dot{m} = r\ V\ A$$

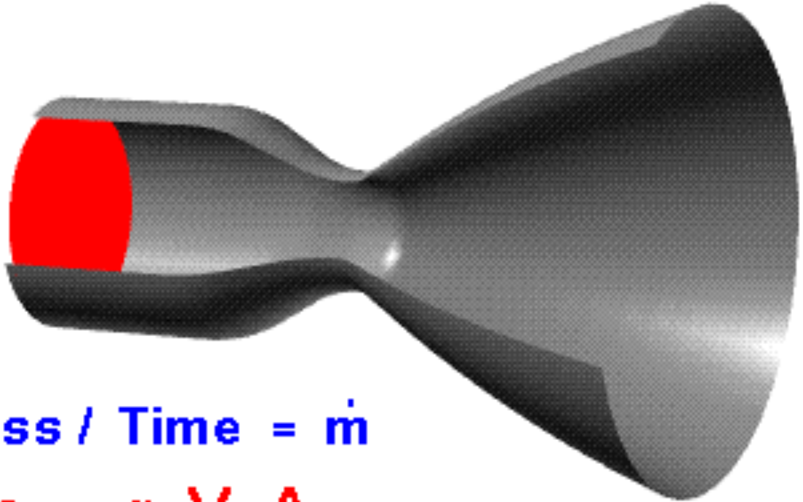
Units Check: $\dfrac{\text{mass}}{\text{length}^3}\ \dfrac{\text{length}}{\text{time}}\ \text{length}^2 = \dfrac{\text{mass}}{\text{time}}$

Continuity : $r\ V\ A = \text{Constant}$

r = Density

V = Velocity

A = Area



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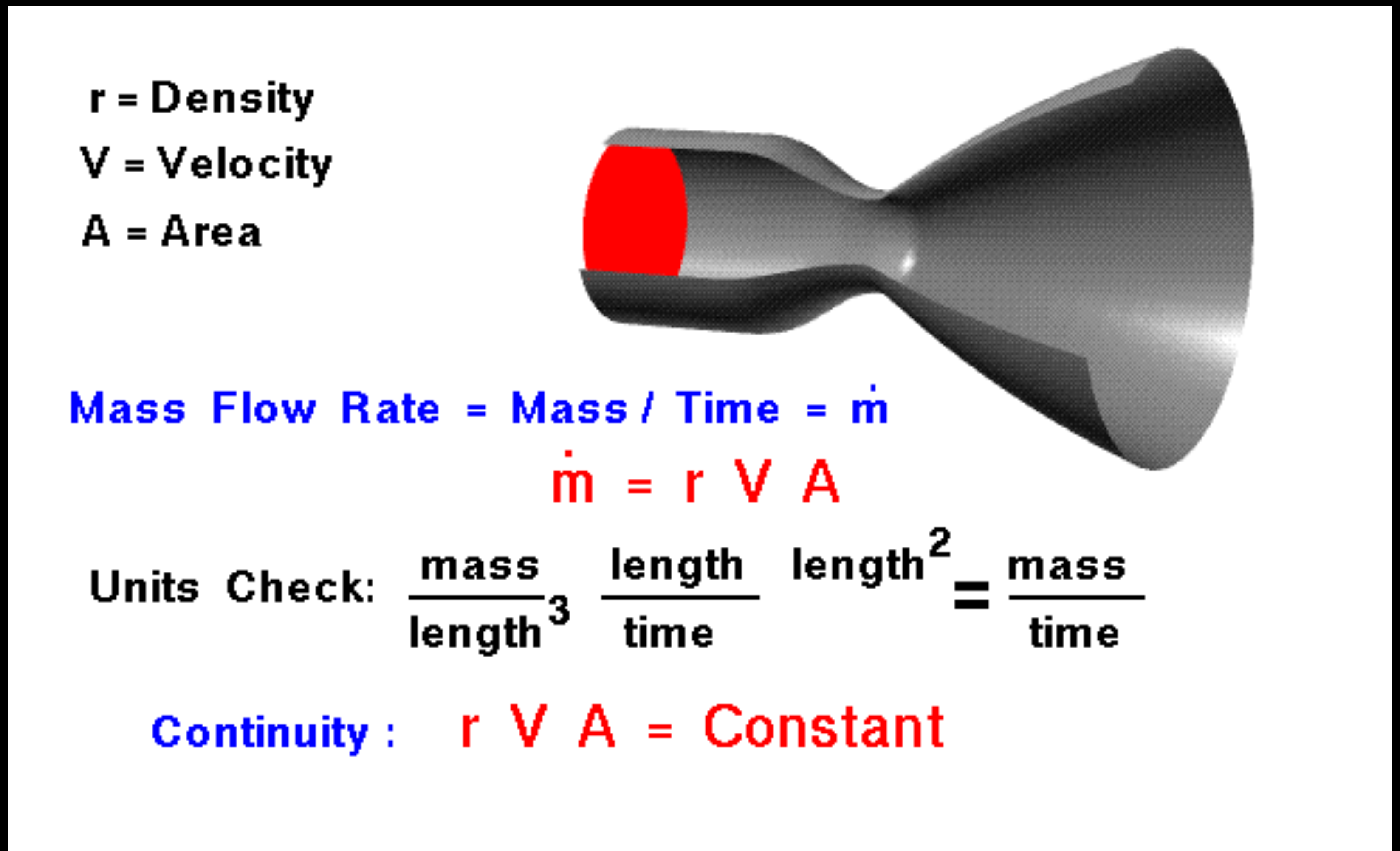
Continuity Equation

- Continuity equation: $\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0$
- A **mass conservation constraint** on Benamou-Brenier formulation
- Ensures probability mass neither created nor destroyed
  (like marginal constraints in Monge/Kantorovich, but at every time step)
- **Analogy**: incompressible fluid flow
- **Intuition**: flow without 'leaks'



Example: 1D Dirac Measures

- Transport $\alpha = \delta_0$ to $\beta = \delta_1$